

Engineering Mathematics

Previous Year Questions • 1998–2026

NUMERICAL METHODS

(EC, EE, IN, CS, ME, CE, CH, PI)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Numerical Methods (EC/EE/IN/CS/ME/CE/CH/PI), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:





Easy



Medium



Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Branch Snapshot

A Maths unit can span up to eight branches (EC, EE, IN, CS, ME, CE, CH, PI), each weighted very differently in the real exam. Right after every Unit divider, a **Branch Snapshot** page lists every branch that unit covers, in the same order the questions themselves follow, each in its own compact box:

Questions	the real total for that branch.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy branch showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Branch Snapshot*

Branch: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real EC questions on this unit; on average EC has drawn about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** for your branch points to a consistently high-yield topic worth prioritizing; a low one is largely someone else's problem. A wide **Range** usually means the topic's importance has shifted over the years, so it is worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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I

Numerical Methods

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   b) Bisection and Secant Method
   c) Newton-Raphson Method
2. Numerical Solution of Linear Equations
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3. Numerical Integration:
   a) Trapezoidal Rule
   b) Simpson's 1/3 Rule and Simpson's 3/8 Rule
4. Numerical Solution of Differential Equations:
   a) Euler's Method
   b) Backward (Implicit) Euler Method
   c) Modified Euler Method (2nd Order Runge-Kutta)
5. Interpolation:
   a) Newton's Forward Difference Method and Backward Difference Method
   b) Newton's Divided Difference Method
   c) Lagrange Interpolation Formula

Note to Reader

Numerical methods are present in branches with paper codes AG, BM, BT, CE, CH, IN, TF, PI, PE, MT, MN, ME according to GATE 2027 syllabus. Before starting please check with the latest syllabus.

Branches: EC,EE,IN,ME,CE,CH,PI

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 8 **MCQ:** 4 **MSQ:** 0 **NAT:** 4
1M: 4 **2M:** 4 **Desc:** 0 **Avg Weightage** ~1M **Range:** 0M(2015)–2M(2010)

EE — Electrical Engineering

Questions: 9 **MCQ:** 7 **MSQ:** 0 **NAT:** 2
1M: 6 **2M:** 3 **Desc:** 0 **Avg Weightage** ~1M **Range:** 0M(2014)–2M(2023)

IN — Instrumentation Engineering

Questions: 4 **MCQ:** 4 **MSQ:** 0 **NAT:** 0
1M: 4 **2M:** 0 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2014)–1M(2014)

ME — Mechanical Engineering

Questions: 21 **MCQ:** 9 **MSQ:** 0 **NAT:** 12
1M: 10 **2M:** 11 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2025)–4M(2023)

CE — Civil Engineering

Questions: 26 **MCQ:** 15 **MSQ:** 0 **NAT:** 11
1M: 10 **2M:** 16 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2025)–4M(2015)

CH — Chemical Engineering

Questions: 31 **MCQ:** 20 **MSQ:** 0 **NAT:** 11
1M: 12 **2M:** 19 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2026)–5M(2016)

PI — Production & Industrial Engineering

Questions: 10 **MCQ:** 8 **MSQ:** 0 **NAT:** 2
1M: 6 **2M:** 4 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2016)–2M(2014)

**See preface for how Avg Weightage and Range are calculated.*

Electronics & Communication Engineering

Q7.17.1 NAT 2017 2M

Starting with $x = 1$, the solution of the equation $x^3 + x = 1$, after two iterations of Newton-Raphson's method (up to two decimal places) is _____

Q7.16.1 NAT 2016 2M

Consider the first order initial value problem $y' = y + 2x - x^2$, $y(0) = 1$, $(0 \le x < \infty)$ with exact solution $y(x) = x^2 + e^x$. For $x = 0.1$, the percentage difference between the exact solution and the solution obtained using a single iteration of the second-order Runge-Kutta method with step-size $h = 0.1$ is _____

Q7.16.2 NAT 2016 2M

The ordinary differential equation $\frac{dx}{dt} = -3x + 2$, with $x(0) = 1$ is to be solved using the forward Euler method. The largest time step that can be used to solve the equation without making the numerical solution unstable is _____

Q7.15.1 NAT 2015 1M

The Newton-Raphson method is used to solve the equation $f(x) = x^3 - 5x^2 + 6x - 8 = 0$. Taking the initial guess as $x = 5$, the solution obtained at the end of the first iteration is _____ (rounded off to two decimal places).

Q7.14.1 MCQ 2014 1M

Match the application to appropriate numerical method:

Applications

$P1$ Numerical integration

$P2$ Solution to a transcendental equation

$P3$ Solution to a system of linear equations

$P4$ Solution to a differential equation

Numerical Method

$M1$ Newton-Raphson Method

$M2$ Runge-Kutta Method

$M3$ Simpson's 1/3-rule

$M4$ Gauss Elimination Method

A. $P1-M3,\ P2-M2,\ P3-M4,\ P4-M1$

B. $P1-M3,\ P2-M1,\ P3-M4,\ P4-M2$

C. $P1-M4,\ P2-M1,\ P3-M3,\ P4-M2$

D. $P1-M2,\ P2-M1,\ P3-M3,\ P4-M4$

Q7.11.1 MCQ 2011 1M

A numerical solution of the equation $f(x) = x + \sqrt{x} - 3 = 0$ can be obtained using Newton-Raphson method. If the starting value is $x = 2$ for the iteration, then the value of x that is to be used in the next step is

A. 0.306 **B.** 0.739 **C.** 1.694 **D.** 2.306

Q7.10.1 MCQ 2010 2M

Consider a differential equation $\frac{dy(x)}{dx} - y(x) = x$ with initial condition $y(0) = 0$. Using Euler's first order method with a step size of 0.1, then the value of $y(0.3)$ is

A. 0.01 **B.** 0.031 **C.** 0.0631 **D.** 0.1

Q7.08.1 MCQ 2008 1M

The recursion relation to solve $x = e^{-x}$ using Newton-Raphson method is

A. $x_{n+1} = e^{-x_n}$

B. $x_{n+1} = x_n - e^{-x_n}$

C. $x_{n+1} = \frac{(1 + x_n)e^{-x_n}}{(1 + e^{-x_n})}$

D. $x_{n+1} = \frac{x_n^2 - e^{-x_n}(1 + x_n) - 1}{x_n - e^{-x_n}}$

Electrical Engineering

Q7.23.1 MCQ 2023 1M

One million random numbers are generated from a statistically stationary process with a Gaussian distribution with mean zero and standard deviation σ_0. The σ_0 is estimated by randomly drawing out 10, 000 numbers of samples (x_n). The estimates $\widehat{\sigma}_1, \widehat{\sigma}_2$ are computed in the following two ways:

$$\widehat{\sigma}_1^2 = \frac{1}{10000}\sum_{n=1}^{10000} x_n^2 \qquad \widehat{\sigma}_2^2 = \frac{1}{9999}\sum_{n=1}^{10000} x_n^2$$

Which of the following statements is true?

A. $E(\widehat{\sigma}_2^2) = \sigma_0^2$

B. $E(\widehat{\sigma}_2) = \sigma_0$

C. $E(\widehat{\sigma}_1^2) = \sigma_0^2$

D. $E(\widehat{\sigma}_1) = E(\widehat{\sigma}_2)$

Q7.23.2 NAT 2023 1M

Three points in the x-y plane are $(-1, 0.8)$, $(0, 2.2)$ and $(1, 2.8)$. The value of the slope of the best fit straight line in the least square sense is _____ (Round off to 2 decimal places).

Q7.15.1 MCQ 2015 1M

If a continuous function $f(x)$ does not have a root in the interval $[a, b]$, then which one of the following statements is TRUE?

A. $f(a) \cdot f(b) = 0$

B. $f(a) \cdot f(b) < 0$

C. $f(a) \cdot f(b) > 0$

D. $f(a)/f(b) \leq 0$

Q7.11.1 MCQ 2011 2M

Solution, the variable x_1 and x_2 for the following equations is to be obtained by employing the Newton-Raphson iteration method:

$$10x_2 \sin x_1 - 0.8 = 0$$
$$10x_2^2 - 10x_2 \cos x_1 - 0.6 = 0$$

Assuming the initial values $x_1 = 0.0$ and $x_2 = 1.0$, the Jacobian matrix is

A. $\begin{bmatrix} 10 & -0.8 \\ 0 & -0.6 \end{bmatrix}$ **B.** $\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$

C. $\begin{bmatrix} 0 & -0.8 \\ 10 & -0.6 \end{bmatrix}$ **D.** $\begin{bmatrix} 10 & 0 \\ 10 & -10 \end{bmatrix}$

Q7.14.1 NAT 2014 1M

The function $f(x) = e^x - 1$ is to be solved using Newton-Raphson method. If the initial value of x_0 is taken 1.0, then the absolute error observed at 2nd iteration is _____

Q7.09.1 MCQ 2009 1M

Let $x^2 - 117 = 0$. The iterative steps for the solution using Newton-Raphson's method is given by

A. $x_{k+1} = \frac{1}{2}\left(x_k + \frac{117}{x_k}\right)$

B. $x_{k+1} = x_k - \frac{117}{x_k}$

C. $x_{k+1} = x_k - \frac{x_k}{117}$

D. $x_{k+1} = x_k - \frac{1}{2}\left(x_k + \frac{117}{x_k}\right)$

Q7.08.1 MCQ 2008 1M

Equation $e^x - 1 = 0$ is required to be solved using Newton's method with an initial guess $x_0 = -1$. Then after one step of Newton's method estimate x_1 of the solution will be given by

A. 0.71828 **B.** 0.36784 **C.** 0.20587 **D.** 0.0000

Q7.08.2 MCQ 2008 2M

A differential equation $\frac{dx}{dt} = e^{-2t}u(t)$ has to be solved using trapezoidal rule of integration with a step size $h = 0.01$ sec. Function $u(t)$ indicates a unit step function. If $x(0) = 0$ then the value of x at $t = 0.01$ sec will be given by

A. 0.00099 **B.** 0.00495 **C.** 0.0099 **D.** 0.0198

Q7.98.2 MCQ 1998 2M

The value of $\int_1^2 \frac{1}{x}\,dx$ computed using Simpson's rule with a step size of $h = 0.25$ is

A. 0.69430 **B.** 0.69385 **C.** 0.69325 **D.** 0.69415

Instrumentation Engineering

Q7.14.1 MCQ 2014 1M

The iteration step in order to solve for the cube roots of a given number 'N' using the Newton-Raphson's method is

A. $x_{k+1} = x_k + \frac{1}{3}\left(N - x_k^3\right)$

B. $x_{k+1} = \frac{1}{3}\left(2x_k + \frac{N}{x_k^2}\right)$

C. $x_{k+1} = x_k - \frac{1}{3}\left(N - x_k^3\right)$

D. $x_{k+1} = \frac{1}{3}\left(2x_k - \frac{N}{x_k^2}\right)$

Q7.13.1 MCQ 2013 1M

While numerically solving the differential equation $\frac{dy}{dx} + 2xy^2 = 0, y(0) = 1$ using Euler's predictor corrector (improved Euler-Cauchy) method with a step size of 0.2, the value of y after the first step is

A. 1.00 **B.** 1.03 **C.** 0.97 **D.** 0.96

Q7.08.1 MCQ 2008 1M

It is known that two roots of the non-linear equation $x^3 - 6x^2 + 11x - 6 = 0$ are 1 and 3. The third root will be

A. j **B.** -j **C.** 2 **D.** 4

Q7.07.1 MCQ 2007 1M

Identify the Newton-Raphson iteration scheme for finding the square root of 2

A. $x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right)$

B. $x_{n+1} = \frac{1}{2}\left(x_n - \frac{2}{x_n}\right)$

C. $x_{n+1} = \frac{2}{x_n}$

D. $x_{n+1} = \sqrt{2 + x_n}$

Mechanical Engineering

Q7.26.1 MCQ 2026 1M

Newton-Raphson method for solving algebraic equations is based on

A. Taylor series

B. Fourier series

C. Laurent series

D. Power series

Q7.26.2 MCQ 2026 1M

The exact solution of $\int_0^4 \frac{dx}{1+x}$ is represented as n. If m represents numerically evaluated value of the above integral using Trapezoidal rule by considering four equal subintervals in the range of x, then $(m - n)$ is

A. 0.074 **B.** −0.074 **C.** −0.003 **D.** 0.003

Q7.25.1 NAT 2025 1M

The values of a function f obtained for different values of x are shown in the table below.

x	0	0.25	0.5	0.75	1.0
$f(x)$	0.9	2.0	1.5	1.8	0.4

Using Simpson's one-third rule,

$$\int_0^1 f(x)\,dx \approx _____$$

[Rounded off to 2 decimal places]

Q7.24.1 MCQ 2024 1M

In order to numerically solve the ordinary differential equation $dy/dt = -y$ for $t > 0$, with an initial condition $y(0) = 1$, the following scheme is employed:

$$\frac{y_{n+1} - y_n}{\Delta t} = -\frac{1}{2}(y_{n+1} + y_n).$$

Here, Δt is the time step and $y_n = y(n\Delta t)$ for $n = 0, 1, 2, \ldots$. This numerical scheme will yield a solution with non-physical oscillations for $\Delta t > h$. The value of h is

A. $\frac{1}{2}$ **B.** 1 **C.** $\frac{3}{2}$ **D.** 2

Q7.23.1 NAT 2023 2M

The smallest perimeter that a rectangle with area of 4 square units can have is _____ units.

[Answer in integer]

Q7.23.2 NAT 2023 2M

The initial value problem

$$\frac{dy}{dt} + 2y = 0, y(0) = 1$$

is solved numerically using the forward Euler's method with a constant and positive time step of Δt.

Let y_n represent the numerical solution obtained after n steps. The condition $|y_{n+1}| \le |y_n|$ is satisfied if and only if Δt does not exceed _____.

[Answer in integer]

Q7.17.1 MCQ 2017 2M

$P(0, 3), Q(0.5, 4)$, and $R(1, 5)$ are three points on the curve defined by $f(x)$. Numerical integration is carried out using both Trapezoidal rule and Simpson's rule within limits $x = 0$ and $x = 1$ for the curve. The difference between the two results will be

A. 0 **B.** 0.25 **C.** 0.5 **D.** 1

Q7.16.1 MCQ 2016 1M

The root of the function $f(x) = x^3 + x - 1$ obtained after first iteration on application of Newton-Raphson scheme using an initial guess of $x_0 = 1$ is

A. 0.682 **B.** 0.686 **C.** 0.750 **D.** 1.000

Q7.16.2 MCQ 2016 1M

Numerical integration using trapezoidal rule gives the best result for a single variable function, which is

A. linear **B.** parabolic

C. logarithmic **D.** hyperbolic

Q7.16.3 NAT 2016 2M

Gauss-Seidel method is used to solve the following equations (as per the given order).

$$x_1 + 2x_2 + 3x_3 = 5$$

$$2x_1 + 3x_2 + x_3 = 1$$

$$3x_1 + 2x_2 + x_3 = 3$$

Assuming initial guess as $x_1 = x_2 = x_3 = 0$, the value of x_3 after the first iteration is _____.

Q7.16.4 NAT 2016 2M

The error in numerically computing the integral $\int_0^{\pi} (\sin x + \cos x)\, dx$ using the trapezoidal rule with three intervals of equal length between 0 and π is _____.

Q7.16.5 NAT 2016 2M

Solve the equation $x = 10\cos(x)$ using the Newton-Raphson method. The initial guess is $x = \frac{\pi}{4}$. The value of the predicted root after the first iteration, up to second decimal, is _____.

Q7.15.1 NAT 2015 2M

The values of function $f(x)$ at 5 discrete points are given below:

x	0	0.1	0.2	0.3	0.4
f(x)	0	10	40	90	160

Using Trapezoidal rule with step size of 0.1, the value of $\int_0^{0.4} f(x)\, dx$ is _____.

Q7.15.2 NAT 2015 2M

Simpson's $\frac{1}{3}$ rule is used to integrate the function $f(x) = \frac{3}{5}x^2 + \frac{9}{5}$ between $x = 0$ and $x = 1$ using the least number of equal sub-intervals. The value of the integral is _____.

Q7.15.3 NAT 2015 2M

Using a unit step size, the value of integral $\int_1^2 x \ln x\, dx$ by trapezoidal rule is _____.

Q7.14.1 NAT 2014 1M

Using the trapezoidal rule, and dividing the interval of integration into three equal sub-intervals, the definite integral $\int_{-1}^{+1} |x|\, dx$ is _____

Q7.14.2 NAT 2014 1M

The real root of the equation $5x - 2\cos x = 0$ (up to two decimal accuracy) is _____

Q7.14.3 MCQ 2014 2M

Consider an ordinary differential equation $\frac{dx}{dt} = 4t + 4$. If $x = x_0$ at $t = 0$, the increment in x calculated using Runge-Kutta fourth order multi-step method with a step size of $\Delta t = 0.2$ is

A. 0.22 **B.** 0.44 **C.** 0.66 **D.** 0.88

Q7.14.4 NAT 2014 2M

The value of $\int_{2.5}^{4} \ln(x)\, dx$ calculated using the Trapezoidal rule with five sub-intervals is _____.

Q7.13.1 MCQ 2013 1M

Match the CORRECT pairs.

Numerical Integration scheme	Fitting Polynomial order
P. Simpson's 3/8 Rule	1. First
Q. Trapezoidal Rule	2. Second
R. Simpson's 1/3 Rule	3. Third

A. $P - 2,\ Q - 1,\ R - 3$

B. $P - 3,\ Q - 2,\ R - 1$

C. $P - 1,\ Q - 2,\ R - 3$

D. $P - 3,\ Q - 1,\ R - 2$

Q7.11.1 MCQ 2011 1M

The integral $\int_1^3 \frac{1}{x}\,dx$ when evaluated by using Simpson's $1/3^{rd}$ rule on two equal sub intervals each of length 1, equals to

A. 1.000 **B.** 1.008 **C.** 1.1111 **D.** 1.120

Civil Engineering

Q7.26.1 NAT 2026 2M

Starting with the first approximation as $x = 0.5$, the second approximation for the root of the following function by the Newton-Raphson method is _____ (rounded off to two decimal places).

$$f(x) = e^{-x} - x$$

Q7.26.2 NAT 2026 2M

Values of y for different values of x are tabulated below.

x	-2	1	2
y	28	4	16

If a second-degree interpolating polynomial $P_2(x)$ is used to represent y, the value of $P_2(0)$ is _____ (rounded off to the nearest integer).

Q7.26.3 MCQ 2026 1M

A fifth-degree polynomial in x is defined for $x > 0$. All coefficients of the polynomial are positive. The first derivative of the polynomial is obtained numerically at a point by using the first-order forward as well as the first-order backward difference methods. Identical step lengths are used for both the methods.

Following statements are made.

(I) Forward difference method underestimates the true derivative.

(II) Backward difference method overestimates the true derivative.

Which one of the following options is CORRECT?

A. Both statements (I) and (II) are FALSE.

B. Both statements (I) and (II) are TRUE.

C. Statement (I) is TRUE and statement (II) is FALSE.

D. Statement (I) is FALSE and statement (II) is TRUE.

Q7.25.1 NAT 2025 2M

Consider the differential equation given below. Using the Euler method with the step size (h) of 0.5, the value of y at $x = 1.0$ is equal to _____ (rounded off to 1 decimal place).

$$\frac{dy}{dx} = y + 2x - x^2; y(0) = 1 \quad (0 \le x < \infty)$$

Q7.24.1 MCQ 2024 1M

The second derivative of a function f is computed using the fourth-order Central Divided Difference method with a step length h. The CORRECT expression for the second derivative is

A. $\frac{1}{12h^2}[-f_{i+2} + 16f_{i+1} - 30f_i + 16f_{i-1} - f_{i-2}]$

B. $\frac{1}{12h^2}[f_{i+2} + 16f_{i+1} - 30f_i + 16f_{i-1} - f_{i-2}]$

C. $\frac{1}{12h^2}[-f_{i+2} + 16f_{i+1} - 30f_i + 16f_{i-1} + f_{i-2}]$

D. $\frac{1}{12h^2}[-f_{i+2} - 16f_{i+1} + 30f_i - 16f_{i-1} - f_{i-2}]$

Q7.24.2 NAT 2024 1M

Consider the data of $f(x)$ given in the table.

i	0	1	2
x_i	1	2	3
$f(x_i)$	0	0.3010	0.4771

The value of $f(1.5)$ estimated using second-order Newton's interpolation formula is _____ (rounded off to 2 decimal places).

Q7.23.1 MCQ 2023 2M

A function $f(x)$, that is smooth and convex-shaped between interval (x_l, x_u) is shown in the figure. This function is observed at odd number of regularly spaced points. If the area under the function is computed numerically, then.

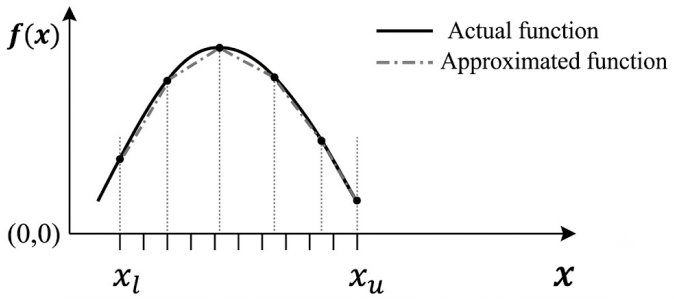


Fig. Q7.23.1

A. the numerical value of the area obtained using the trapezoidal rule will be less than the actual

B. the numerical value of the area obtained using the trapezoidal rule will be more than the actual

C. the numerical value of the area obtained using the trapezoidal rule will be exactly equal to the actual

D. with the given details, the numerical value of area cannot be obtained using trapezoidal rule

Q7.22.1 NAT 2022 1M

Consider the following recursive iteration scheme for different values of variable P with the initial guess $x_1 = 1$:

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{P}{x_n}\right), n = 1, 2, 3, 4, 5$$

For $P = 2$, x_5 is obtained to be 1.414, rounded-off to three decimal places. For $P = 3$, x_5 is obtained to be 1.732, rounded-off to three decimal places. If $P = 10$, the numerical value of x_5 is _____ (round off to three decimal places).

Q7.17.1 NAT 2017 2M

Consider the equation $\frac{du}{dt} = 3t^2+1$ with $u = 0$ at $t = 0$. This is numerically solved by using the forward Euler method with a step size $\Delta t = 2$. The absolute error in the solution at the end of the first time step is _____.

Q7.16.1 NAT 2016 2M

Newton-Raphson method is to be used to find root of equation $3x - e^x + \sin x = 0$. If the initial trial value for the root is taken as 0.333, the next approximation for the root would be _____ (note: answer up to three decimal).

Q7.16.2 MCQ 2016 1M

The quadratic approximation of $f(x) = x^3 - 3x^2 - 5$ at the point $x = 0$ is

A. $3x^2 - 6x - 5$ **B.** $-3x^2 - 5$

C. $-3x^2 + 6x - 5$ **D.** $3x^2 - 5$

Q7.15.1 NAT 2015 2M

For step-size, $\Delta x = 0.4$, the value of following integral using Simpson's 1/3 rule is _____.

$$\int_0^{0.8} \left(0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5\right) dx$$

Q7.15.2 NAT 2015 2M

In Newton-Raphson iterative method, the initial guess value (x_{ini}) is considered as zero while finding the roots of the equation:

$$f(x) = -2 + 6x - 4x^2 + 0.5x^3$$

The correction, Δx, to be added to x_{ini} in the first iteration is _____.

Q7.15.3 MCQ 2015 2M

The integral $\int_{x_1}^{x_2} x^2\, dx$ with $x_2 > x_1 > 0$ is evaluated analytically as well as numerically using a single application of the trapezoidal rule. If I is the exact value of the integral obtained analytically and J is the approximate value obtained using the trapezoidal rule, which of the following statements is correct about their relationship?

A. J > I

B. J < I

C. J = I

D. Insufficient data to determine the relationship

Q7.15.4 NAT 2015 2M

The quadratic equation $x^2 - 4x + 4 = 0$ is to be solved numerically, starting with the initial guess $x_0 = 3$. The Newton-Raphson method is applied once to get a new estimate and then the Secant method is applied once using in the initial guess and this new estimate. The estimated value of the root after the application of the Secant method is _____.

Q7.13.1 NAT 2013 2M

There is no value of x that can simultaneously satisfy both the given equations. Therefore, find the 'least squares error' solution to the two equations, i.e., find the value of x that minimizes the sum of squares of the errors in the two equations:

$$2x = 3$$

$$4x = 1$$

Q7.12.1 MCQ 2012 1M

The estimate of $\int_{0.5}^{1.5} \frac{dx}{x}$ obtained using Simpson's rule with three-point function evaluation exceeds the exact value by

A. 0.235 **B.** 0.068 **C.** 0.024 **D.** 0.012

Q7.11.1 MCQ 2011 2M

The square root of a number N is to be obtained by applying the Newton – Raphson iteration to the equation $x^2 - N = 0$. If i denotes the iteration index, the correct iterative scheme will be

A. $x_{i+1} = \frac{1}{2}\left[x_i + \frac{N}{x_i}\right]$ **B.** $x_{i+1} = \frac{1}{2}\left[x_i^2 - \frac{N}{x_i^2}\right]$

C. $x_{i+1} = \frac{1}{2}\left[x_i + \frac{N^2}{x_i}\right]$ **D.** $x_{i+1} = \frac{1}{2}\left[x_i - \frac{N}{x_i}\right]$

Q7.10.1 MCQ 2010 2M

The table below gives values of a function $f(x)$ obtained for values of x at intervals of 0.25

X	0	0.25	0.50	0.75	1
F(x)	1	0.9412	0.88	0.64	0.5

The value of the integral of the function between the limits 0 to 1, using Simpson's rule is

A. 0.7854 **B.** 2.3562 **C.** 3.1416 **D.** 7.5000

Q7.09.1 MCQ 2009 2M

The area under the curve shown between $x = 1$ and $x = 3$ is to be evaluated using the trapezoidal rule. The following points on the curve are given

Point	X – coordinate (m)	Y – coordinate (m)
1	1	1
2	2	4
3	3	9

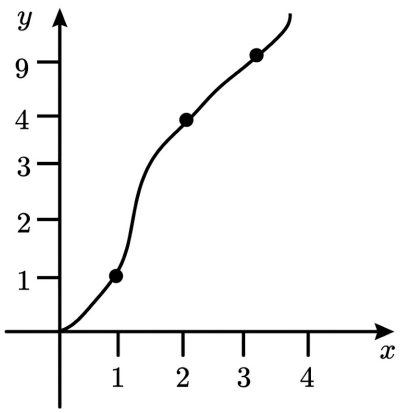


Fig. Q7.09.1

The evaluated area (in m^2) will be

A. 7 **B.** 8.67 **C.** 9 **D.** 18

Q7.08.1 MCQ 2008 1M

The Newton-Raphson iteration $x_{n+1} = \frac{1}{2}\left(x_n + \frac{R}{x_n}\right)$ can be used to compute

A. square of R **B.** reciprocal of R

C. square root of R **D.** logarithm of R

Q7.08.2 MCQ 2008 2M

If the interval of integration is divided into two equal intervals of width 1.0, the value of the definite integral $\int_1^3 \log_e x\, dx$ using simpson's one – third rule will be

A. 0.5 **B.** 0.80 **C.** 1.00 **D.** 0.29

Q7.07.1 MCQ 2007 1M

The following equation needs to be numerically solved using the Newton – Raphson method $x^3 + 4x - 9 = 0$. The iterative equation for this purpose is (k indicates the iteration level)

A. $X_{k+1} = \frac{2X_k^3 + 9}{3X_k^2 + 4}$ **B.** $X_{k+1} = \frac{3X_k^3 + 9}{2X_k^2 + 9}$

C. $X_{k+1} = X_k - 3_k^2 + 4$ **D.** $X_{k+1} = \frac{4X_k^2+3}{9X_k^2+2}$

Q7.07.2 MCQ 2007 1M

Given that one root of the equation $x^3 - 10x^2 + 31x - 30 = 0$ is 5 then other roots are

A. 2 and 3 **B.** 2 and 4

C. 3 and 4 **D.** -2 and -3

Q7.05.1 MCQ 2005 2M

Given $a > 0$, we wish to calculate its reciprocal value $\frac{1}{a}$ by using Newton - Raphson method for $f(x) = 0$. For $a = 7$ and starting with $x_0 = 0.2$ the first two iterations will be

A. 0.11, 0.1299 **B.** 0.12, 0.1392

C. 0.12, 0.1416 **D.** 0.13, 0.1428

Q7.05.2 MCQ 2005 1M

Given $a > 0$, we wish to calculate its reciprocal value $\frac{1}{a}$ by using Newton - Raphson method for $f(x) = 0$. The Newton - Raphson algorithm for the function will be

A. $X_{k+1} = \frac{1}{2}\left(X_k + \frac{a}{X_k}\right)$ **B.** $X_{k+1} = X_k + \frac{a}{2}X_k^2$

C. $X_{k+1} = 2X_k - aX_k^2$ **D.** $X_{k+1} = 2X_k - \frac{a}{2}X_k^2$

Chemical Engineering

Q7.26.1 NAT 2026 1M

Using single step trapezoidal rule, the value of

$$\int_0^{0.5} (1 + 16x - 20x^2)\, dx$$

is _____ (rounded off to two decimal places).

Q7.25.1 NAT 2025 2M

The Newton-Raphson method is used to find the root of

$$f(x) = x^2 - x - 1 = 0$$

Starting with an initial guess $x(0) = 1$, the second iterate $x(2)$ is

_____ (rounded off to 2 decimal places).

Q7.24.1 NAT 2024 2M

The Newton-Raphson method is used to solve $f(x) = 0$, where $f(x) = e^x - 5x$. If the initial guess $x(0) = 1.0$, the value of the next iterate, $x(1)$, rounded off to 2 decimal places, is

_____

Q7.24.2 MCQ 2024 1M

The first non-zero term in the Taylor series expansion of $(1 - x) - e^{-x}$ about $x = 0$ is

A. 1 **B.** -1 **C.** $\frac{x^2}{2}$ **D.** $-\frac{x^2}{2}$

Q7.23.1 MCQ 2023 2M

Simpson's one-third rule is used to estimate the definite integral

$$I = \int_{-1}^{1} \sqrt{(1 - x^2)}\, dx$$

with an interval length of 0.5. Which one of the following is the CORRECT estimate of I obtained using this rule?

A. $\frac{1}{3} - \frac{1}{\sqrt{3}}$ **B.** $\frac{1}{3} + \frac{2}{\sqrt{3}}$ **C.** $\frac{1}{3} + \frac{1}{\sqrt{3}}$ **D.** $\frac{1}{3} - \frac{2}{\sqrt{3}}$

Q7.22.1 NAT 2022 2M

The equation $\frac{dy}{dx} = xy^2 + 2y + x - 4.5$ with the initial condition $y(x = 0) = 1$ is to be solved using a predictor-corrector approach. Use a predictor based on the implicit Euler's method and a corrector based on the trapezoidal rule of integration, each with a full-step size of 0.5. Considering only positive values of y, the value of y at $x = 0.5$ is _____(rounded off to three decimal places).

Q7.21.1 MCQ 2021 1M

An ordinary differential equation (ODE), $\frac{dy}{dx} = 2y$, with an initial condition $y(0) = 1$, has the analytical solution $y = e^{2x}$. Using Runge-Kutta second order method, numerically integrate the ODE to calculate y at $x = 0.5$ using a step size of $h = 0.5$. If the relative percentage error is defined as,

$$\varepsilon = \left| \frac{y_{\text{analytical}} - y_{\text{numerical}}}{y_{\text{analytical}}} \right| \times 100$$

then the value of ε at $x = 0.5$ is

A. 0.06 **B.** 0.8 **C.** 4.0 **D.** 8.0

Q7.21.2 MCQ 2021 1M

The function $\cos(x)$ is approximated using Taylor series around $x = 0$ as $\cos(x) \approx 1 + a\,x + b\,x^2 + c\,x^3 + d\,x^4$. The values of a, b, c and d are

A. $a = 1, \quad b = -0.5, \quad c = -1, \quad d = -0.25$

B. $a = 0, \quad b = -0.5, \quad c = 0, \quad d = 0.042$

C. $a = 0, \quad b = 0.5, \quad c = 0, \quad d = 0.042$

D. $a = -0.5, \quad b = 0, \quad c = 0.042, \quad d = 0$

Q7.21.3 NAT 2021 2M

To solve an algebraic equation $f(x) = 0$, an iterative scheme of the type $x_{n+1} = g(x_n)$ is proposed, where $g(x) = x - \frac{f(x)}{f'(x)}$. At the solution $x = s$, $g'(s) = 0$ and $g''(s) \neq 0$. The order of convergence for this iterative scheme near the solution is _____

Q7.20.1 MCQ 2020 1M

Which one of the following methods requires specifying an initial interval containing the root (i.e., bracketing) to obtain the solution of $f(x) = 0$, where $f(x)$ is a continuous non-linear algebraic function?

A. Newton-Raphson method

B. regula falsi method

C. secant method

D. fixed point iteration method

Q7.20.2 NAT 2020 2M

Consider the following dataset.

x	1	3	5	15	25
$f(x)$	6	8	10	12	5

The value of the integral $\int_1^{25} f(x)\,dx$ using Simpson's $1/3^{rd}$ rule is _____(round off to 1 decimal place).

Q7.19.1 NAT 2019 2M

The Newton-Raphson method is used to determine the root of the equation $f(x) = e^{-x} - x$. If the initial guess for the root is 0, the estimate of the root after two iterations is _____ (rounded off to three decimal places).

Q7.18.1 MCQ 2018 1M

The fourth order Runge-Kutta (RK4) method to solve an ordinary differential equation $\frac{dy}{dx} = f(x, y)$ is given as

$$y(x+h) = y(x) + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = hf(x, y)$$

$$k_2 = hf\left(x + \frac{h}{2}, y + \frac{k_1}{d}\right)$$

$$k_3 = hf\left(x + \frac{h}{2}, y + \frac{k_2}{d}\right)$$

$$k_4 = hf(x + h, y + k_3)$$

For a special case when the function f depends solely on x, the above RK4 method reduces to

A. Euler's explicit method

B. Trapezoidal rule

C. Euler's implicit method

D. Simpson's 1/3 rule

Q7.17.1 NAT 2017 1M

The number of positive roots of the function $f(x)$ shown below in the range $0 < x < 6$ is

_____

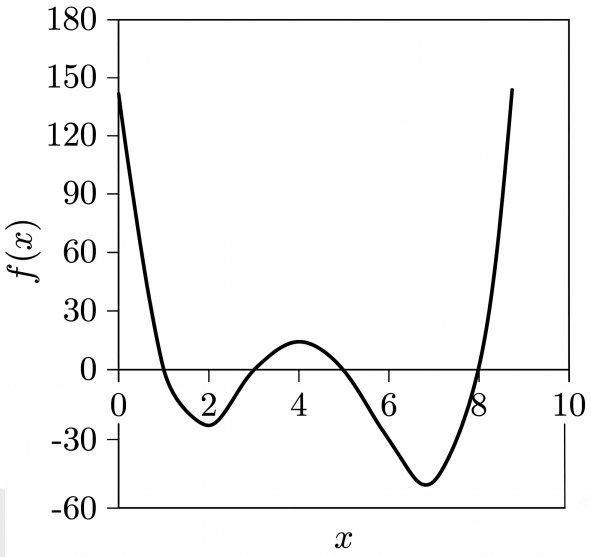


Fig. Q7.17.1

Q7.17.2 MCQ 2017 2M

Match the problem type in Group-1 with the numerical method in Group-2.

Group-1	**Group-2**
P) System of linear algebraic equations	I) Newton-Raphson
Q) Non-linear algebraic equations	II) Gauss-Seidel
R) Ordinary differential equations	III) Simpson's rule
S) Numerical integration	IV) Runge-Kutta

A. P-II, Q-I, R-III, S-IV **B.** P-I, Q-II, R-IV, S-III

C. P-IV, Q-III, R-II, S-I **D.** P-II, Q-I, R-IV, S-III

Q7.16.1 MCQ 2016 1M

Which one of the following is an iterative technique for solving a system of simultaneous linear algebraic equations?

A. Gauss elimination **B.** Gauss-Jordan

C. Gauss-Seidel **D.** LU decomposition

Q7.16.2 NAT 2016 2M

The model $y = mx^2$ is to be fit to the data given below.

x	1	$\sqrt{2}$	$\sqrt{3}$
y	2	5	8

Using linear regression, the value (rounded off to the second decimal place) of m is

_____

Q7.16.3 NAT 2016 2M

Values of $f(x)$ in the interval $[0, 4]$ are given below.

x	0	1	2	3	4
$f(x)$	3	10	21	36	55

Using Simpson's 1/3 rule with a step size of 1, the numerical approximation (rounded off to the second decimal place) of $\int_0^4 f(x)\,dx$ is

---

Q7.15.1 MCQ 2015 2M

The solution of the non-linear equation $x^3 - x = 0$ is to be obtained using Newton-Raphson method. If the initial guess is $x = 0.5$, the method converges to which one of the following values?

A. -1 **B.** 0 **C.** 1 **D.** 2

Q7.13.1 NAT 2013 2M

The value of the integral $\int_{0.1}^{0.5} e^{-x^3}\,dx$ evaluated by Simpson's rule using 4 subintervals (up to 3 digits after the decimal point) is

---

Q7.12.1 MCQ 2012 2M

The Newton-Raphson method is used to find the roots of the equation

$$f(x) = x - \cos \pi x \quad 0 \le x \le 1$$

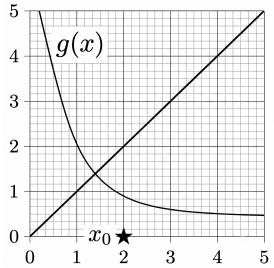
If the initial guess for the root is 0.5, then the value of x after the first iteration is

A. 1.02 **B.** 0.62 **C.** 0.55 **D.** 0.38

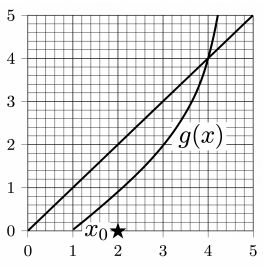
Q7.11.1 MCQ 2011 2M

In the fixed point iteration method for solving equations of the form $x = g(x)$, the $(n+1)^{\text{th}}$ iteration value is $x_{n+1} = g(x_n)$, where x_n represents the nth iteration value. $g(x)$ and corresponding initial guess value x_0 in the domain of interest are shown in the following choices. Which ONE of these choices leads to a converged solution for x?

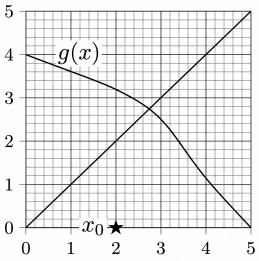
A.



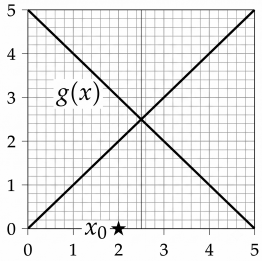
B.



C.



D.



Q7.10.1 MCQ 2010 1M

A root of the equation $x^4 - 3x + 1 = 0$ needs to be found using the Newton-Raphson method. If the initial guess, x_0, is taken as 0, then the new estimate x_1, after the first iteration is

A. $\frac{1}{3}$ **B.** $-\frac{1}{3}$ **C.** 3 **D.** -3

Q7.09.1 MCQ 2009 2M

Using the trapezoidal rule and 4 equal intervals ($n = 4$), the calculated value of the integral (rounded to the first place of decimal) $\int_0^{\pi} \sin\theta\,d\theta$ is

A. 1.7 **B.** 1.9 **C.** 2.0 **D.** 2.1

Q7.08.1 MCQ 2008 1M

A nonlinear function $f(x)$ is defined in the interval $-1.2 < x < 4$ as illustrated in the figure below. The equation $f(x) = 0$ is solved for x within this interval by using the Newton-Raphson iterative scheme. Among the initial guesses (I_1, I_2, I_3 and I_4), the guess that is likely to lead to the root most rapidly is

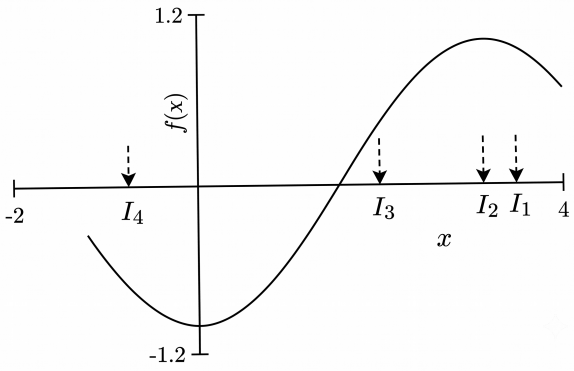


Fig. Q7.08.1

A. I_1 **B.** I_2 **C.** I_3 **D.** I_4

Q7.08.2 MCQ 2008 2M

Using Simpson's 1/3 rule and FOUR equally spaced intervals ($n = 4$), estimate the value of the integral $\int_0^{\pi/4} \frac{\sin x}{\cos^3 x}\, dx$

A. 0.3887 **B.** 0.4384 **C.** 0.5016 **D.** 0.5527

Q7.08.3 MCQ 2008 2M

The following differential equation is to be solved numerically by the Euler's explicit method.

$$\frac{dy}{dx} = x^2 y - 1.2y \quad \text{with } y(0) = 1$$

A step size of 0.1 is used. The solution for y at $x = 0.1$ is

A. 0.880 **B.** 0.905 **C.** 1.000 **D.** 1.100

Q7.06.1 MCQ 2006 1M

The ordinary differential equation $dY/dt = f(Y)$ is solved using the approximation $Y(t + \Delta t) = Y(t) + f[Y(t)]/\Delta t$. The numerical error introduced by the approximation at each step is

A. proportional to Δt

B. proportional to $(\Delta t)^2$

C. independent of Δt

D. proportional to $(1/\Delta t)$

Q7.06.2 MCQ 2006 1M

The trepezoidal rule of integration when applied to $\int_a^b f(x)\, dx$ will give the exact value of the integral

A. if $f(x)$ is a linear function of x

B. if $f(x)$ is a quadratic function of x

C. for any $f(x)$

D. for no $f(x)$

Q7.06.3 MCQ 2006 2M

The Newton-Raphson method is used to solve the equation, $(x-1)^2 + x - 3 = 0$. The method will fail in the very first iteration if the initial guess is

A. 0 **B.** 0.5 **C.** 1 **D.** 3

Q7.05.1 MCQ 2005 2M

The function $f(x)$ satisfies the equation $f(x) = 0$ at $x = x_e$. The Newton-Raphson iterative method converges to the solution in one step, regardless of the initial guess, if

A. $f(x)$ is a linear function of x

B. $f(x)$ is a quadratic function of x

C. $f(x)$ is a cubic function of x

D. $f(x)$ is an exponential function of x

Production & Industrial Engineering

Q7.16.1 MCQ 2016 1M

To solve the equation $2\sin x = x$ by Newton-Raphson method, the initial guess was chosen to be $x = 2.0$. Consider x in radian only. The value of x (in radian) obtained after one iteration will be closest to

A. −8.101 **B.** 1.901 **C.** 2.099 **D.** 12.101

Q7.15.1 MCQ 2015 2M

In numerical integration using Simpson's rule, the approximating function in the interval is a

A. constant **B.** straight line

C. cubic B-Spline **D.** Parabola

Q7.14.1 NAT 2014 1M

Using the Simpson's $1/3^{rd}$ rule, the value of $\int y\, dx$ computed, for the data given below, is _____.

x	1	3	5
y	2	6	4

Q7.14.2 NAT 2014 1M

If the equation $\sin(x) = x^2$ is solved by Newton Raphson's method with the initial guess of $x = 1$, then the value of x after 2 iterations would be _____.

Q7.10.1 MCQ 2010 2M

The following algorithm computes the integral $J = \int_a^b f(x)\, dx$ from the given values $f_j = f(x_j)$ at equidistant points $x_0 = a$, $x_1 = x_0 + h$, $x_2 = x_0 + 2h, \ldots x_{2m} = x_0 + 2mh = b$ compute

$$S_0 = f_0 + f_{2m}$$

$$S_1 = f_1 + f_3 + \cdots + f_{2m-1}$$

$$S_2 = f_2 + f_4 + \cdots + f_{2m-2}$$

$$J = \frac{h}{3}\left[S_0 + 4(S_1) + 2(S_2)\right]$$

The rule of numerical integration, which uses the above algorithm is

A. Rectangle rule **B.** Trapezoidal

C. Four – point rule **D.** Simpson's rule

Q7.09.1 MCQ 2009 1M

During the numerical solution of a first order differential equation using the Euler (also known as Euler Cauchy) method with step size h, the local truncation error is of the order of

A. h^2 **B.** h^3 **C.** h^4 **D.** h^5

Q7.07.1 MCQ 2007 2M

Matching exercise choose the correct one out of the alternatives A, B, C, D

Group – I

P. 2nd order differential equations

Q. Non-linear algebraic equations

R. Linear algebraic equations

S. Numerical integration

Group – II

(1) Runge – Kutta method

(2) Newton – Raphson method

(3) Gauss Elimination

(4) Simpson's Rule

A. $P-3, Q-2, R-4, S-1$

B. $P-2, Q-4, R-3, S-1$

C. $P-1, Q-2, R-3, S-4$

D. $P-1, Q-3, R-2, S-4$

Q7.05.1 MCQ 2005 2M

The real root of the equation $xe^x = 2$ is evaluated using Newton – Raphson's method. If the first approximation of the value of x is 0.8679, the 2nd approximation of the value of x correct to three decimal places is

A. 0.865 **B.** 0.853 **C.** 0.849 **D.** 0.838

Q7.05.2 MCQ 2005 1M

For solving algebraic and transcendental equation which one of the following is used?

A. Coulomb's theorem

B. Newton - Raphson method

C. Euler's method

D. Stoke's theorem

Q7.05.3 MCQ 2005 1M

Newton – Raphson formula to find the roots of an equation $f(x) = 0$ is given by

A. $X_{n+1} = X_n - \dfrac{f(X_n)}{f^1(X_n)}$ **B.** $X_{n+1} = X_n + \dfrac{f(X_n)}{f^1(X_n)}$

C. $X_{n+1} = \dfrac{f(X_n)}{X_n f^1(X_n)}$ **D.** $X_{n+1} = \dfrac{X_n f(X_n)}{f^1(X_n)}$

Answer Key

ELECTRONICS & COMMUNICATION ENGINEERING					
Q7.17.1 0.68 to 0.70	Q7.16.1 0.06	Q7.16.2 0.66	Q7.15.1 4.29	Q7.14.1 B	Q7.11.1 C
Q7.10.1 B	Q7.08.1 C				
ELECTRICAL ENGINEERING					
Q7.23.1 C	Q7.23.2 0.99 to 1.01	Q7.15.1 C	Q7.11.1 B	Q7.14.1 0.06	Q7.09.1 A
Q7.08.1 A	Q7.08.2 C	Q7.98.2 C			
INSTRUMENTATION ENGINEERING					
Q7.14.1 B	Q7.13.1 D	Q7.08.1 C	Q7.07.1 A		
MECHANICAL ENGINEERING					
Q7.26.1 A	Q7.26.2 A	Q7.25.1 1.61 to 1.65	Q7.24.1 D	Q7.23.1 7.999 to 8.001	Q7.23.2 0.999 to 1.001
Q7.17.1 A	Q7.16.1 C	Q7.16.2 A	Q7.16.3 -6	Q7.16.4 0.186	Q7.16.5 1.56
Q7.15.1 22	Q7.15.2 2	Q7.15.3 0.69	Q7.14.1 1.1	Q7.14.2 0.3726	Q7.14.3 D
Q7.14.4 1.7532	Q7.13.1 D	Q7.11.1 C			
CIVIL ENGINEERING					
Q7.26.1 0.55 to 0.58	Q7.26.2 2	Q7.26.3 A	Q7.25.1 2.4 to 2.8	Q7.24.1 A	Q7.24.2 0.16 to 0.18
Q7.23.1 A	Q7.22.1 3.1 to 3.2	Q7.17.1 8	Q7.16.1 0.3601	Q7.16.2 B	Q7.15.1 1.367
Q7.15.2 0.33	Q7.15.3 A	Q7.15.4 2.33	Q7.13.1 0.5	Q7.12.1 D	Q7.11.1 A
Q7.10.1 A	Q7.09.1 C	Q7.08.1 C	Q7.08.2 D	Q7.07.1 A	Q7.07.2 A
Q7.05.1 B	Q7.05.2 C				
CHEMICAL ENGINEERING					
Q7.26.1 1.25	Q7.25.1 1.63	Q7.24.1 -0.01 to 0.01	Q7.24.2 D	Q7.23.1 B	Q7.22.1 0.875
Q7.21.1 D	Q7.21.2 B	Q7.21.3 2	Q7.20.1 B	Q7.20.2 241.4 to 242.4	Q7.19.1 0.565 to 0.567

Q7.18.1 D	Q7.17.1 3	Q7.17.2 D	Q7.16.1 C	Q7.16.2 2.57	Q7.16.3 94.67
Q7.15.1 A	Q7.13.1 0.385	Q7.12.1 D	Q7.11.1 A	Q7.10.1 A	Q7.09.1 B
Q7.08.1 C	Q7.08.2 C	Q7.08.3 A	Q7.06.1 B	Q7.06.2 A	Q7.06.3 B
Q7.05.1 A					
PRODUCTION & INDUSTRIAL ENGINEERING					
Q7.16.1 B	Q7.15.1 D	Q7.14.1 20	Q7.14.2 0.86 to 0.89	Q7.10.1 D	Q7.09.1 A
Q7.07.1 C	Q7.05.1 B	Q7.05.2 B	Q7.05.3 A		

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